

Communication Systems

Module 2 & Module 3 — Easy-to-Read In-Sem Revision Notes

Focus: AM, DSB-SC, SSB, FM, Carson's Rule and Modulators

MODULE 2 — AMPLITUDE MODULATION

1. Amplitude Modulation (AM)

Definition: AM is the process in which the amplitude of a high-frequency carrier is varied according to the instantaneous amplitude of the message signal, while carrier frequency and phase remain constant.

Message: $v_m = V_m \cos(2\pi f_m t)$

Carrier: $v_c = V_c \cos(2\pi f_c t)$

AM wave: $v(t) = V_c [1 + m \cos(2\pi f_m t)] \cos(2\pi f_c t)$

Components: Carrier f_c , USB = $f_c + f_m$, LSB = $f_c - f_m$.

Bandwidth: $BW = 2f_m$.

2. Modulation Index of AM

$$m = V_m / V_c$$

From the envelope: $m = (V_{\max} - V_{\min}) / (V_{\max} + V_{\min})$

m < 1: under-modulation **m = 1:** 100% modulation **m > 1:** over-modulation/distortion.

3. Power in AM

$$P_c = V_c^2 / (2R)$$

$$P_{\text{USB}} = P_{\text{LSB}} = (m^2/4)P_c$$

$$P_{\text{SB}} = (m^2/2)P_c$$

$$\text{Total power: } P_T = P_c (1 + m^2/2)$$

4. AM Generation

Low-level modulation: modulation is performed at low power, followed by RF amplification. Suitable for low-power transmitters.

High-level modulation: message is applied at the collector/high-power stage. Suitable for high-power AM transmitters.

5. Envelope Detector — AM Demodulation

Purpose: Recover the original message from an AM wave.

Components: diode, resistor and capacitor.

Working: diode rectifies the AM signal; capacitor follows the envelope; the RC network filters the carrier and produces the recovered message.

RC too small: ripple distortion. **RC too large:** diagonal clipping. **Correct RC:** proper envelope detection.

6. DSB-SC

Definition: Double Sideband Suppressed Carrier transmits both sidebands while suppressing the carrier.

$$v(t) \propto \cos(2\pi f_m t) \cos(2\pi f_c t)$$

Using the product identity, the output contains components at $f_c + f_m$ and $f_c - f_m$.

Bandwidth: $2f_m$.

7. Balanced Modulator

Purpose: Generate DSB-SC by cancelling the carrier.

Principle: It effectively multiplies the message and carrier signals. Carrier components cancel, leaving the two sidebands.

8. Ring Modulator

Components: four diodes and two transformers.

Working: the carrier acts as a switching signal; the circuit alternately passes the message signal; carrier components cancel and sidebands remain.

9. SSB

Definition: Single Sideband transmits only one sideband—USB or LSB—while suppressing the carrier and the other sideband.

Bandwidth: $BW = f_m$.

Generation methods: filter method, phase-shift method and Weaver method.

10. Filter Method of SSB

Block sequence: Message → Balanced Modulator → Band-Pass Filter → SSB output.

Working: first generate DSB-SC; then use a band-pass filter to remove one sideband.

11. VSB

Definition: Vestigial Sideband transmits one complete sideband and a small portion of the other sideband.

Application: television transmission.

Quick Comparison — AM Systems

System	Carrier	Sidebands	Bandwidth
DSBFC	Present	2	$2f_m$
DSBSC	Suppressed	2	$2f_m$
SSB	Suppressed	1	f_m
VSB	Present/controlled	1 + part	Slightly $> f_m$

MODULE 3 — ANGLE MODULATION

1. Frequency Modulation (FM)

Definition: FM is the process in which the frequency of the carrier varies according to the message signal, while the carrier amplitude remains constant.

FM equation: $v(t) = V_c \cos[2\pi f_c t + \beta \sin(2\pi f_m t)]$

Here β is the FM modulation index.

2. Frequency Deviation

Definition: Maximum change in carrier frequency from its unmodulated value.

$$\Delta f = k_f V_m$$

3. FM Modulation Index

$$\beta = \Delta f / f_m$$

where Δf is maximum frequency deviation and f_m is the message frequency.

4. Instantaneous Frequency

$$f_i = f_c + \Delta f \cos(2\pi f_m t)$$

Instantaneous phase: $\theta(t) = 2\pi f_c t + \beta \sin(2\pi f_m t)$

5. Carson's Rule

Purpose: Estimate the practical bandwidth of an FM signal.

Formula: $BW = 2(\Delta f + f_m)$

For multiple message frequencies, use the maximum frequency deviation and highest modulating frequency.

6. Narrowband FM (NBFM)

Condition: $\beta \ll 1$

Features: small bandwidth, small frequency deviation, and behaviour approximately similar to AM.

7. Wideband FM (WBFM)

Condition: $\beta \gg 1$

Features: larger bandwidth and frequency deviation, with improved noise immunity.

Application: FM radio broadcasting.

8. Direct FM Generation

Definition: The oscillator frequency is directly varied by the message signal.

Examples: reactance modulator and varactor-diode modulator.

9. Varactor Diode FM Modulator

Principle: A varactor diode acts as a voltage-controlled capacitor.

Working: message voltage changes diode capacitance $\rightarrow$ resonant frequency changes $\rightarrow$ FM is produced.

Advantages: simple operation and good frequency stability.

10. Reactance Modulator (JFET)

Principle: A JFET circuit behaves like a variable reactance.

Working: message voltage changes the gate voltage → effective reactance changes → oscillator frequency changes → FM is generated.

11. Indirect FM — Armstrong Method

Principle: Generate phase modulation and use frequency multipliers to obtain the required FM.

Steps: integrate/message-process as required → phase modulation → frequency multiplication.

Advantage: high frequency stability.

Direct vs Indirect FM

Feature	Direct FM	Indirect FM
Frequency stability	Lower	Higher
Circuit complexity	Simpler	More complex
Principle	Directly vary oscillator frequency	Use phase modulation + frequency multiplication

IMPORTANT NUMERICALS

1. Carson's Rule Example

Given: $\Delta f = 75 \text{ kHz}$, $f_m = 15 \text{ kHz}$

$$BW = 2(75 + 15) = \mathbf{180 \text{ kHz}}$$

2. FM Modulation Index Example

Given: $\Delta f = 50 \text{ kHz}$, $f_m = 10 \text{ kHz}$

$$\beta = 50/10 = \mathbf{5}$$

EXAM QUICK REVISION

- **Module 2:** AM equation, modulation index, AM power, envelope detector, balanced modulator, DSB-SC, SSB generation, VSB.
- **Module 3:** FM equation, frequency deviation, modulation index, Carson's Rule, NBFM/WBFM, varactor modulator, reactance modulator, Armstrong method.
- **Numericals:** Practise AM power, modulation index, bandwidth and Carson's Rule.

Exam tip: For 5-mark theory questions, write the definition → equation → labelled block diagram/working → key points/application. For numericals, write Given → Formula → Substitution → Answer with units.